

Joey Sehan Back

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EDUCATION

Carnegie Mellon University

Master of Science in Robotics

Pittsburgh, PA

Aug. 2026 – Present

- Fulbright New Zealand Science & Innovation Graduate Award

University of Auckland

Bachelor of Engineering (Honours) in Computer Systems

Auckland, New Zealand

Mar. 2022 – Nov. 2025

- First Class Honours; Dean's Honours List 2024, 2025; Exchange semester @ **Seoul National University**
- Co-authored robotics papers published in RO-MAN (in press), HRI and ICSR conferences

SKILLS

Programming Languages: Python, Java, C, C++, VHDL, Typescript

Software & Tools: Git, Arduino, ESP32, VS Code, PyCharm, Quartus, LTspice

Languages: English, Korean

EXPERIENCE

Serato LTD

Test Analyst

Auckland, New Zealand

Mar. 2026 – Jul. 2026

- Led AI adoption in QA, improving testing productivity

Software Engineer Intern

Nov. 2025 – Feb. 2026

- Developed in C++ and XML, implementing new features and performing bug fixes

University of Auckland, CARES Lab

Summer Research Scholar

Auckland, New Zealand

Nov. 2024 – Feb. 2025

- Researched and developed a robot chat system incorporating text, speech, and physical expression generation for human-robot interactions
- Co-authored a paper on an LLM-based multi-modal generation system for universal agent platforms; presented and published at HRI 2025

Cochl Inc

Software Engineer Intern

Seoul, South Korea

Jan. 2024 – Mar. 2024

- Led backend implementation of an RFC standard auth-server, utilizing AWS services
- Studied DNN techniques for sound analysis including feature extraction via FFT and mel spectrogram analysis

University of Auckland, CARES Lab

Undergraduate Research Assistant

Auckland, New Zealand

Aug. 2022 – Nov. 2025

- Developed navigation and HRI systems on in-house, and commercial robot platforms
- Researched and implemented DNN techniques for American Sign Language recognition

CiLab

Robotics Coach

Auckland, New Zealand

Feb. 2021 – Mar. 2026

- Organized and delivered educational programs for 6th through 12th-grade students.
- Mentored the New Zealand International Robot Olympiad team
- Led the volunteering effort for the New Zealand Robot Olympiad for 5 years

PROJECTS

Multi-Robot Role-Based Orchestration System

- Built a Centralized Orchestrator using WebSockets and REST APIs for context-aware multi-robot hand-offs
- Engineered a tri-layered memory system combining key-point extraction with RAG
- Deployed across 5 robotic platforms and validated via user studies ($N = 20$), boosting trust and interaction fluidity

LLM based multi-modal generation for universal agent platforms

- Developed in-house expressive, low-cost robot platforms using 3D printing and efficient servo joint mechanisms
- Built real-time, LLM-driven multi-modal conversation system
- Co-authored a paper and video, published and presented at HRI 2025

LLM based multi-modal generation for universal agent platforms

- Developed in-house expressive, low-cost robot platforms using 3D printing and efficient servo joint mechanisms
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FPGA-Based HMPSoC for Power Frequency Detection

- Developed a heterogeneous Multiprocessor System-on-Chip (HMPSoC) integrating a custom processor, Nios II core, and application-specific processors via a TDMA-based Network-on-Chip for real-time power system frequency detection on an Intel DE1 FPGA board

AI Square

- Conducted LLM finetuning for a no-code web development ChatBot, and integration with the client's existing web development IDE

MENYOY: Mental and Emotional Nurturing of Youth

- Developed a low-cost teddy bear robot for educating ASD affected children through physical interaction
- Presented and awarded at ICSR 2019, HRI 2021, and WRO 2022

Mr TT. (Turtle Toy)

- Turtle plush toy with speech recognition and conversational interaction targeted towards ASD affected children through natural speech and educational games
- Presented and awarded at IRO 2020 and IEEE TENCON 2021

PUBLICATIONS

- Junhu Song, **Joey Sehan Back**, Bruce MacDonald, and Ho Seok Ahn. "Orchestrating Role-Based Social Continuity for Heterogeneous Multi-Robot Teams." Accepted to: *IEEE Int'l Conf. on Robot and Human Interactive Communication (RO-MAN)*. 2026.
- Junhu Song, Minwoo Lee, **Joey Sehan Back**, Peter Cheong, and Ho Seok Ahn. "Role-Adaptive Communication Framework with Large Language Models for Multi-robot Systems." In: *Int'l Conf. on Social Robotics + AI (ICSR+AI)*. 2026.
- **Joey Sehan Back**, Peter Cheong, and Ho Seok Ahn. "Is the Earth Flat? Mythbusting with HRI." In: *Proc. of the ACM/IEEE Int'l Conf. on Human-Robot Interaction (HRI)*. 2025.
- Daniel Sung Hyun Shin, **Joey Sehan Back**, Tony Lee, Peter Cheong, and Ho Seok Ahn. "TUTUS: Track Utilizing Transport for User Safety." In: *Companion Proc. of the ACM/IEEE Int'l Conf. on Human-Robot Interaction (HRI)*. 2021.

ACHIEVEMENTS

IEEE RO-MAN Special session paper (in press)	2026
1st Place, Dosh Case Competition	2025
International Social Robotics Conference (ICSR) - Innovative Solution Award	2025
IEEE RO-MAN Design competition - Best Context Design Award	2025
HRI Human-Robot Interaction Conference Demo and Video Finalist	2025
UoA Highly Commended Research in ECSE	2025
University of Auckland Summer Research Scholarship	2024
1st Place, ZURU FMCG Case Competition	2024
5th Place, Energy Case Competition	2024
Prime Minister's Scholarship for Asia	2023
University of Auckland 360 Exchange Award – Asia	2023
LEGO World Robot Olympiad (5th Place)	2022
IEEE NZ North & A Society Service Award	2022

CLUBS & SOCIETIES

University of Auckland Web Development & Consulting Club

2024 – 2026

- 2026: Project advisor for a team of 10, developing a university event ticketing platform
- 2025: Technical lead for a team of 9 to build a platform for a 2,000+ member volunteering organisation, featuring event tracking, admin tools, social networking, and a public landing page
- 2024: Was a backend developer on the CPNZ webapp for NZ Police community patrol officers, streamlining all administrative duty processes

University of Auckland Case Club

2024 – 2025

INTERESTS

Sports, music, travel, food, personal finance and ranking local lattes

Referees available on request